Sugar Twin: a real-time Digital Twin for Type 2 Diabetes

Predicting glucose spikes 2 hours ahead by fusing EHR + wearable data, for India

[FILL IN: team name] · [FILL IN: college / incubator]

SYNTHETIC DATA ONLY: NOT FOR CLINICAL USE

Post-meal spikes: frequent, invisible, preventable

~101 M

people living with diabetes in India (ICMR-INDIAB study, 2023)

3 months

typical gap between HbA1c checks, which average away daily spikes

1.9/day

sustained spikes above 180 mg/dL per patient-day in our simulated cohort

Why it matters in India

Rice- and wheat-heavy meals, late dinners, a high-waist "thin-fat" phenotype and early onset make post-prandial spikes common. Spikes drive long-term complications, yet CGMs and smartwatches stream data that nobody turns into timely, explainable, personalised action for the patient or the doctor.

Use case: warn 2 h before glucose exceeds 180 mg/dL

1 Spike risk

Calibrated probability of glucose > 180 mg/dL in the next 120 min, refreshed every 5 min

2 2-hour forecast

Glucose at +30 / 60 / 90 / 120 min with a P10–P90 uncertainty band

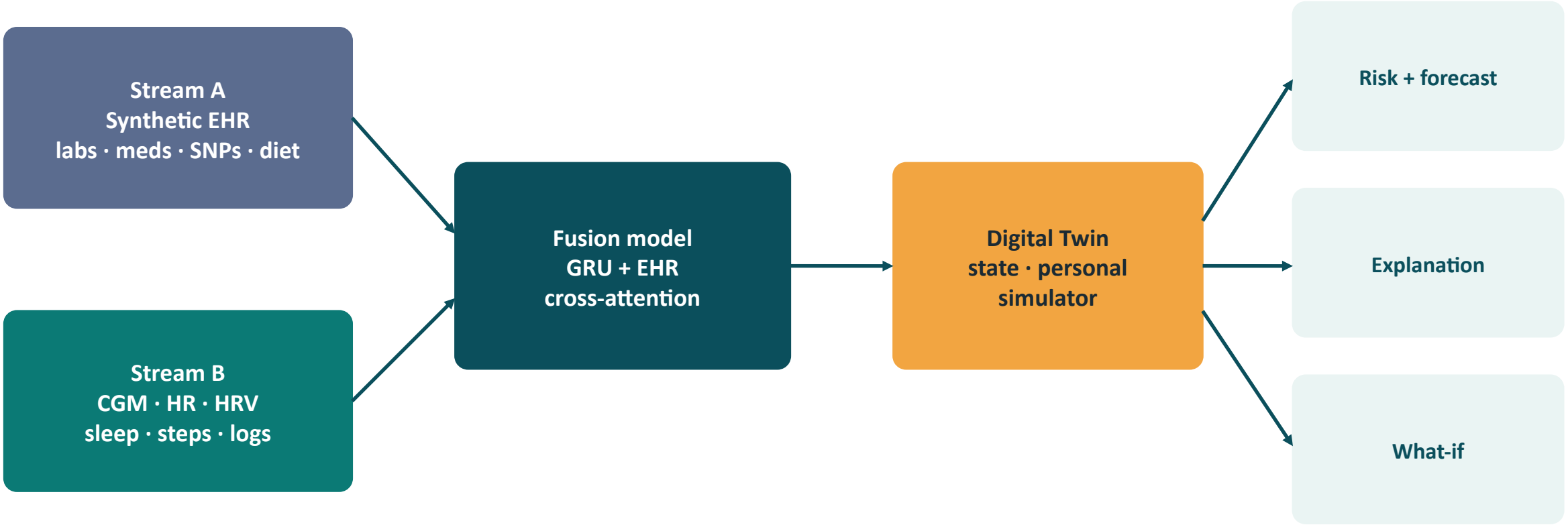
3 Alert level

GREEN / AMBER / RED with a "spike in ~X min" countdown; icon + text, never colour alone

4 Why + what if

Top factors in plain language, and counterfactual simulations of walks, meals, sleep and medication timing

Two data streams, one twin, two role-based apps



Doctor dashboard (/) and patient PWA (/patient/) are two clients of the same twin and the same 5-minute stream

Synthetic patients whose EHR drives their physiology

A Indian T2D EHR generator

Correlated demographics, labs, comorbidities and Indian prescribing (metformin, gliptins, premix insulin). South-Asian BMI (23/25) and waist cut-offs, simulated TCF7L2 / SLC30A8 / KCNQ1 + polygenic score, diet and staple. FHIR R4 export.

150

patients × 30 days at 5-min resolution

0.95

correlation of sensor GMI with EHR HbA1c (the EHR shapes the signals)

B Physiological world simulator

Extended Bergman minimal model: idli, poha, dal-chawal and biryani meals, drug effects, dawn phenomenon, exercise, sleep debt, stress, plus CGM lag, noise, dropouts and compression lows.

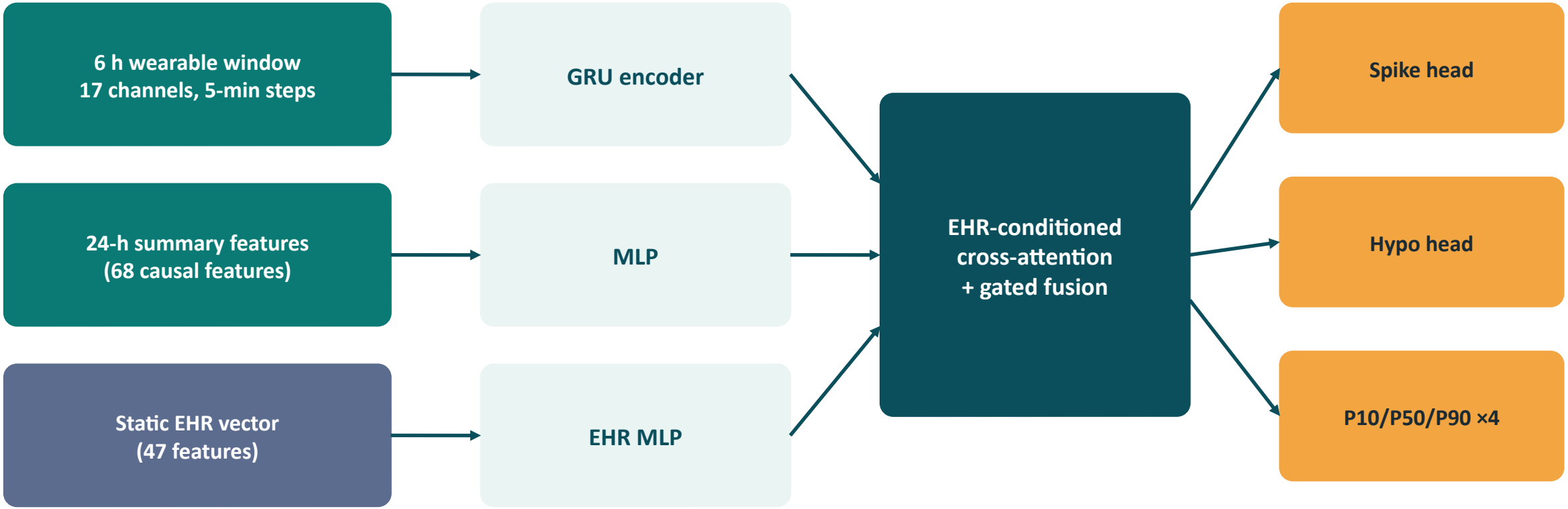
72.9%

time in range 70–180 mg/dL

0

real patients: every assumption is listed in docs/ASSUMPTIONS.md

LightGBM baseline vs multimodal deep fusion



Baseline: LightGBM on the same 118 causal features. Both are isotonic-calibrated on validation patients. Explanations: TreeSHAP / Integrated Gradients.

Evaluation built to avoid leakage and report honestly

1 Split by patient

105 train / 22 validation / 23 test patients; test patients are never used for tuning

2 Temporal hold-out

Last 20% of days of the training patients; label windows never cross the cut-off

3 Cold start

First monitoring day (6–24 h of history), where fusion should matter most

4 Live alert replay

Every 5 min, with a refractory period: lead time and false alerts per patient-day

5 Fairness

Age group, sex, South-Asian BMI band, urban / rural

6 Robustness

Extra CGM noise, 30% CGM dropout, 50% missing watch data, no meal logs

Headline results on unseen test patients

0.938

ROC-AUC (deep fusion)

0.873

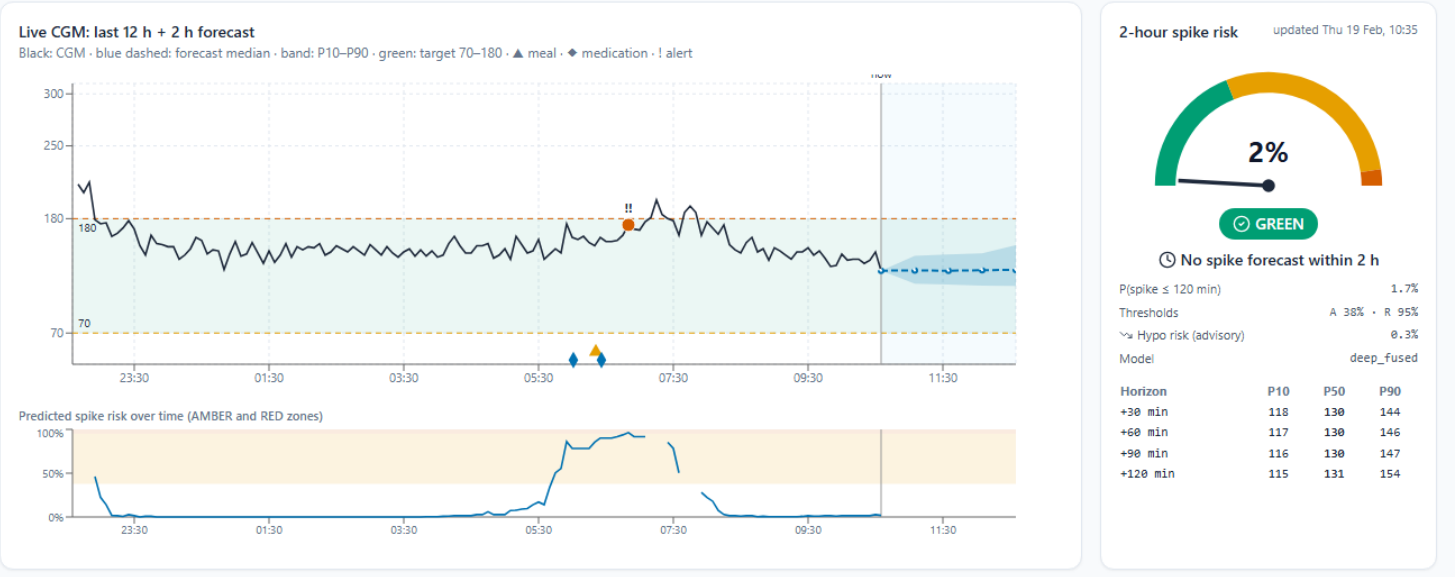
PR-AUC (random = 0.312)

80%

of spikes caught at RED, 1.0 false alert / patient-day

45 min

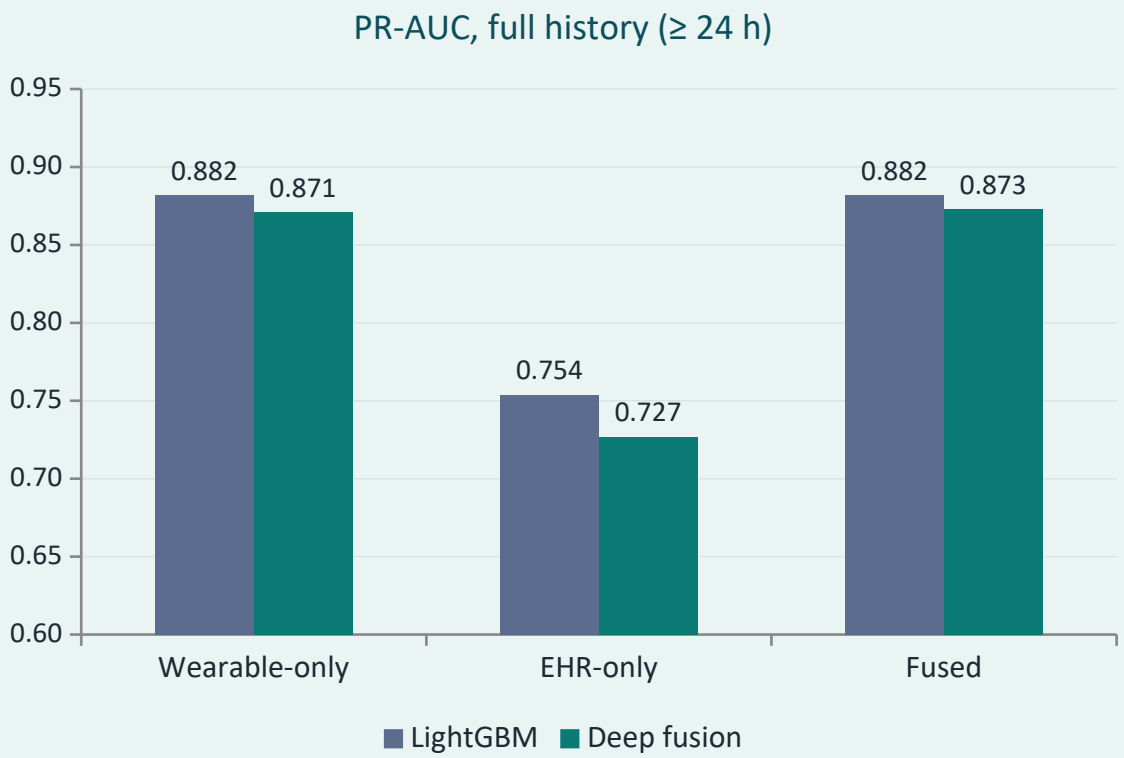
mean warning before the spike at RED (≈85 min at AMBER)

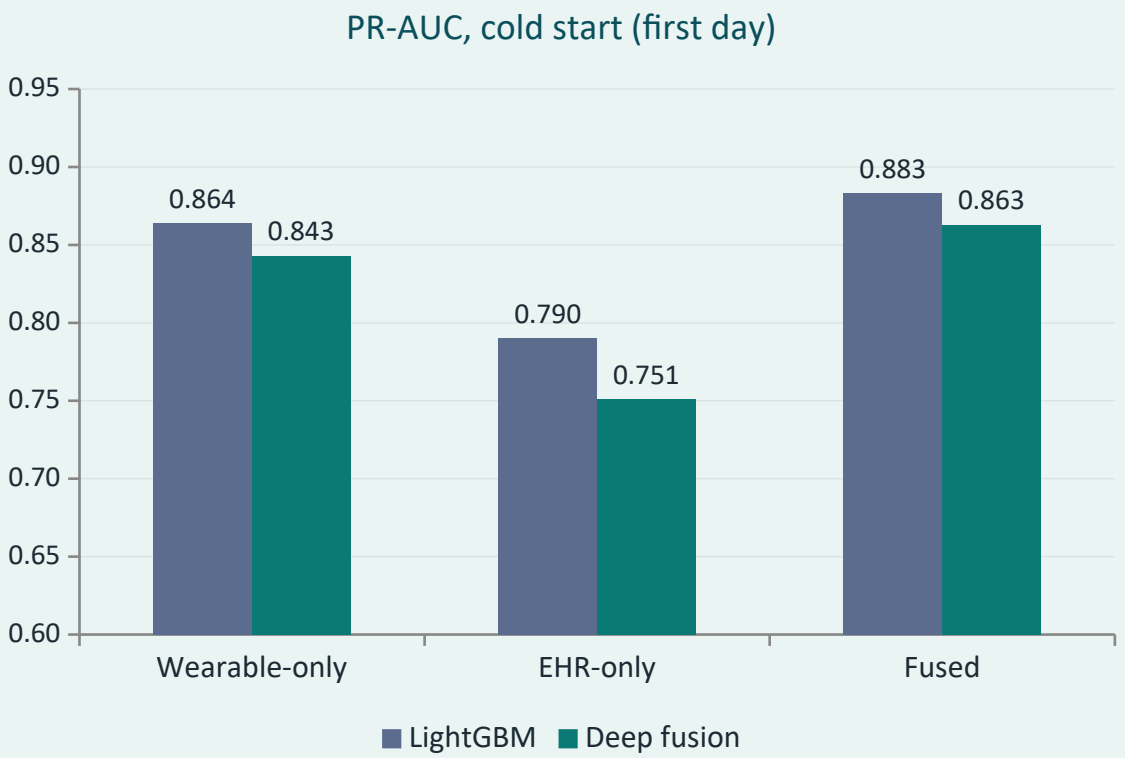


Calibrated and honest

Brier score 0.092. Temporal hold-out ROC-AUC 0.950. Shown left: risk entered RED about 2 h before a breakfast spike.

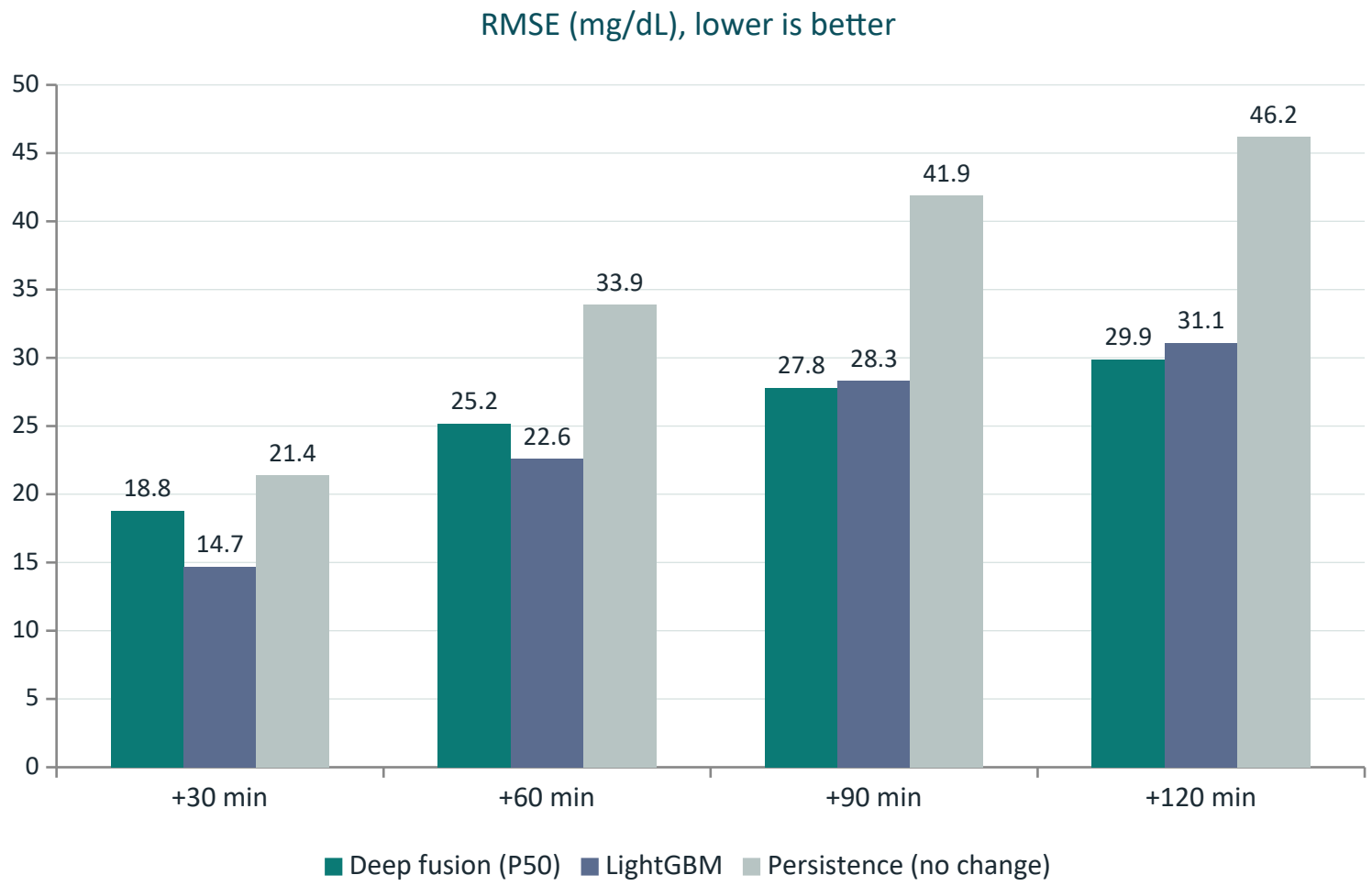
Does fusion help? Not in steady state, but yes at cold start





Fusion gain: steady state +0.002 (deep) · cold start +0.020 (deep), +0.019 (GBM) · at RED: 80% vs 75% of spikes caught

2-hour forecast beats "no change" at every horizon



+60 min

Deep RMSE 25.2 vs 33.9 mg/dL for "no change"

Uncertainty band

P10–P90 covers 71% of outcomes at +60 min (target 80%), slightly over-confident

A personalised simulator that answers "what if?"

1 **Starts from the EHR prior**

It never sees the simulator's hidden truth.

2 **Syncs to every CGM reading**

An observer keeps the model state on track.

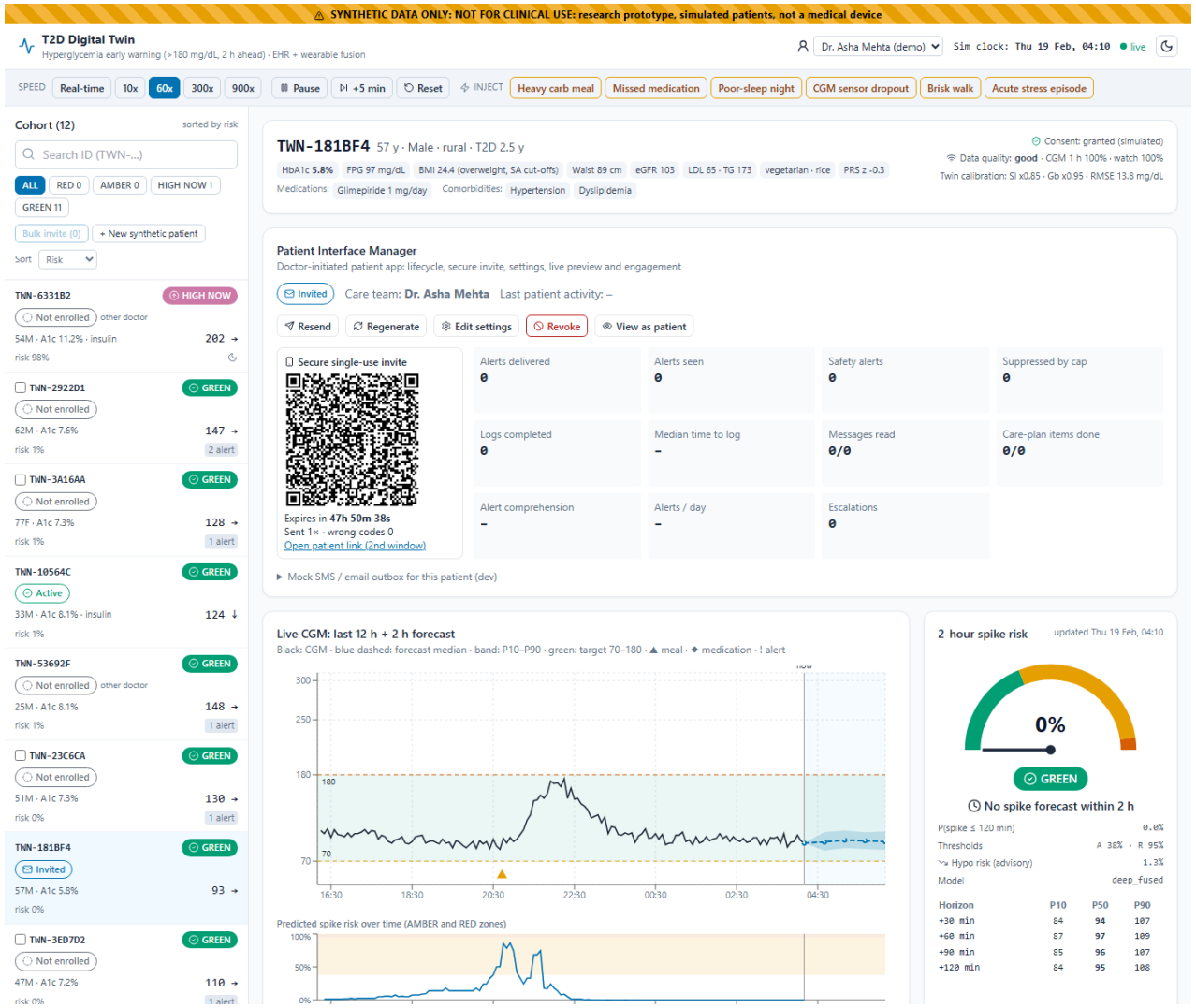
3 **Re-calibrates every 6 h**

Insulin sensitivity and basal set-point fitted to the last 24 h.

Walk 15 min now	*lower peak, less time > 180*
Roti instead of rice	*smaller, slower rise*
Dinner at 19:00 vs 21:00	*peak moves away from bedtime*
Sleep 5 h vs 8 h	*higher next-morning response*

Estimates for discussion: never treatment advice

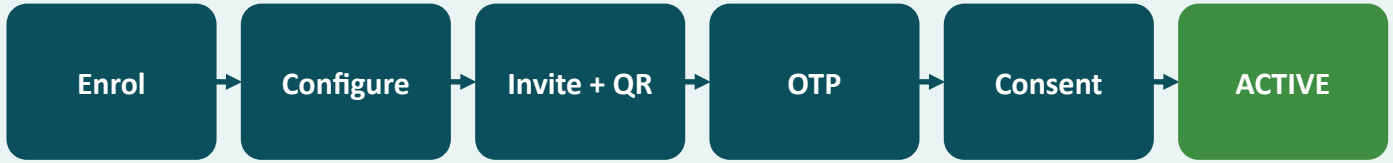
Doctor dashboard: triage, explanation, app control



- Cohort sorted by risk, with a status column and bulk invite
- Live CGM + 2-h forecast band, risk gauge and countdown
- "Why this alert?" factors from EHR and wearable
- Inject events live: heavy meal, missed dose, poor sleep, dropout
- Patient Interface Manager: secure invite + QR, live preview, engagement
- Every action written to an audit trail

Patient app: doctor-launched, in the patient's language





- Home: one plain sentence, glucose + trend, 2-h estimate in the target zone, one next action
- My Day: timeline and "why" in everyday language
- Try it: walk / roti vs rice / earlier dinner
- Quick Log: Indian food presets in under 10 seconds
- Care: alerts, care-plan cards, doctor messages
- Me: progress, access log, withdraw consent

Personal food-impact AI: which foods push my sugar up?



My food patterns

Foods to watch

Chicken curry + rice Watch
After Chicken curry + rice, your sugar rose about 112 mg/dL (peak 241); above 180 after 100% of these meals.
based on 3 meals · some evidence

Sambar rice Watch
After Sambar rice, your sugar rose about 94 mg/dL (peak 229); above 180 after 100% of these meals.
based on 2 meals · some evidence

Dal-chawal Watch
After Dal-chawal, your sugar rose about 98 mg/dL (peak 232); above 180 after 100% of these meals.
based on 5 meals · reliable

For foods marked Watch: try a smaller portion, add dal or vegetables, or walk 15 minutes after eating.

Chai (sugar) + biscuits : Usually fine (peak 178, based on 2 meals)

Millet khichdi : Usually fine (peak 178, based on 2 meals)

Patterns we noticed

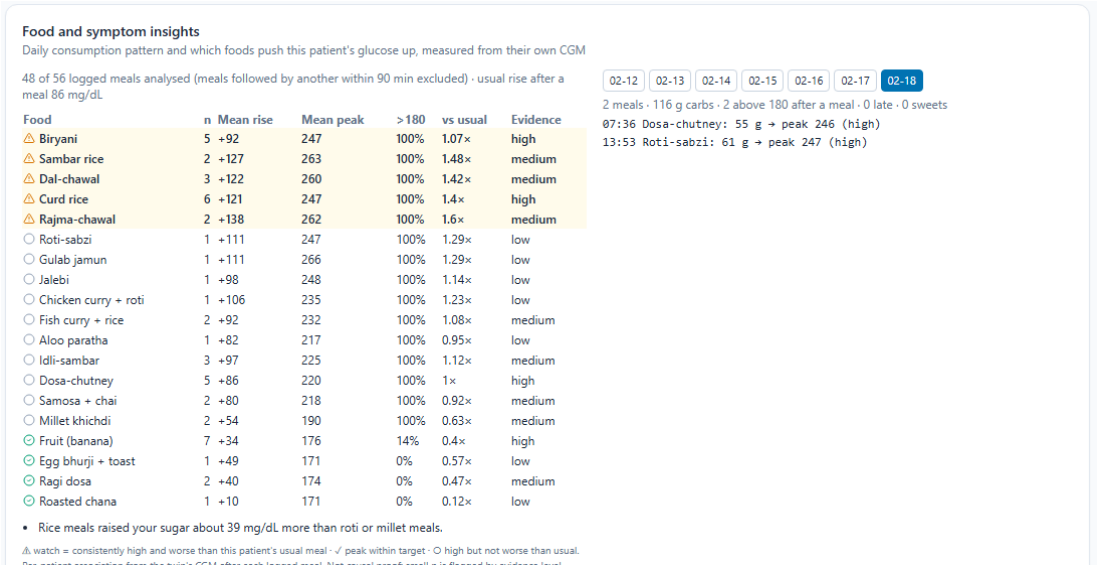
- Rice meals raised your sugar about 33 mg/dL more than roti or millet meals.

Food diary (last 7 days)

14:22 Dal-chawal 80 g carbs · peak 209

	g carbs	high after meal	late meals
02-12	174	2/4	1
02-13	175	2/3	0
02-14	0	0/0	0
02-15	275	2/3	0
02-16	86	0/2	1
02-17	140	1/2	0
02-18	80	1/1	0

Measured from your own sensor after each logged meal. More logged meals give more reliable results.



Food and symptom insights

Daily consumption pattern and which foods push this patient's glucose up, measured from their own CGM

48 of 56 logged meals analysed (meals followed by another within 90 min excluded) · usual rise after a meal 86 mg/dL

02-12 02-13 02-14 02-15 02-16 02-17 02-18

2 meals · 116 g carbs · 2 above 180 after a meal · 0 late · 0 sweets

07:36 Dosa-chutney: 55 g → peak 246 (high)
13:53 Roti-sabzi: 61 g → peak 247 (high)

Food	n	Mean rise	Mean peak	>180	vs usual	Evidence
Biryani	5	+92	247	100%	1.07×	high
Sambar rice	2	+127	263	100%	1.48×	medium
Dal-chawal	3	+122	260	100%	1.42×	medium
Curd rice	6	+121	247	100%	1.4×	high
Rajma-chawal	2	+138	262	100%	1.6×	medium
Roti-sabzi	1	+111	247	100%	1.29×	low
Gulab jamun	1	+111	266	100%	1.29×	low
Jalebi	1	+98	248	100%	1.14×	low
Chicken curry + roti	1	+106	235	100%	1.23×	low
Fish curry + rice	2	+92	232	100%	1.08×	medium
Aloo paratha	1	+82	217	100%	0.95×	low
Idli-sambar	3	+97	225	100%	1.12×	medium
Dosa-chutney	5	+86	220	100%	1×	high
Samosa + chai	2	+80	218	100%	0.92×	medium
Millet khichdi	2	+54	190	100%	0.63×	medium
Fruit (banana)	7	+34	176	14%	0.4×	high
Egg bhurji + toast	1	+49	171	0%	0.57×	low
Ragi dosa	2	+40	174	0%	0.47×	medium
Roasted chana	1	+10	171	0%	0.12×	low

- Rice meals raised your sugar about 39 mg/dL more than roti or millet meals.

⚠ watch = consistently high and worse than this patient's usual meal · ✓ peak within target · ○ high but not worse than usual.

- Measures each meal from the patient's own CGM
- Flags foods worse than this patient's usual meal
- Links logged symptoms to foods eaten 4 h before
- Warns gently when a flagged food is logged again

14 days

food diary per patient, with daily carbs, late meals, sweets and highs after meals

3 demo

patient accounts (English / Hindi / Tamil), already activated for the demo

Safety, privacy and consent built in, not bolted on

1 Always-on safety

RED and low-glucose alerts cannot be disabled or capped; first aid and tap-to-call 108

2 No medication changes

Reminders only for the existing prescription; questions go to the doctor

3 Consent gate

No twin data before consent; withdrawal disconnects instantly and stops logs

4 Secure invites

Signed single-use tokens, expiry, OTP with lockout, device binding, rate limits

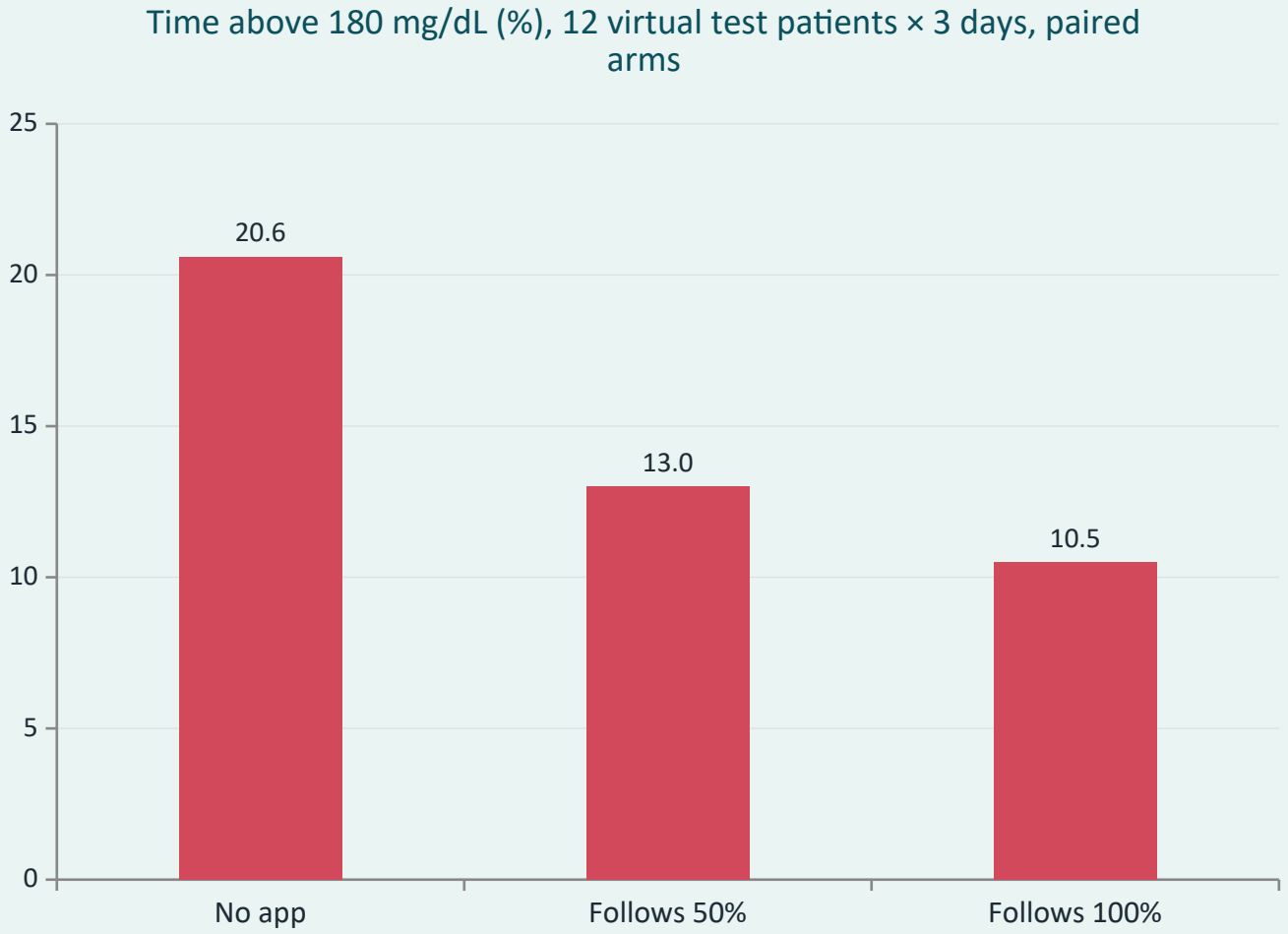
5 Role-based access

Doctors see only assigned patients; patients only themselves; caregiver opt-in

6 Audit + minimisation

Every view and change logged; the patient sees who viewed their data

Simulated patients who follow the app spend less time high



Simulation only

A mechanism check inside our own simulator, not clinical evidence. Real-world benefit needs a trial.

No added lows

Time below 70: 0% → 0%. Spike counts fell by about one per day in both following arms.

Technical stack and engineering quality

Data & ML	Python 3.12 · NumPy / pandas / SciPy · PyTorch (GRU fusion) · LightGBM · scikit-learn · SHAP
Backend	FastAPI · Uvicorn · WebSockets · Pydantic · SQLite audit · PyJWT
Frontend	React 18 · Vite · TypeScript · Tailwind · Recharts · react-i18next · PWA
Ops	make / tasks.py · Docker Compose · training 6 min + evaluation on a laptop CPU

63 + 16

automated tests (pytest + vitest), incl. end-to-end

1 cmd

make demo: backend, doctor dashboard and patient app on one port

Limitations and the path to real-world validation

- Synthetic data: the models partly learn our simulator's assumptions
- Not a medical device; no clinical validation
- Small test set (23 patients); fairness estimates are noisy
- Hypoglycaemia is rare in the cohort, so the hypo head is advisory
- Hindi and Tamil text is machine-drafted: needs native review

1 **Retrospective study**
Consented Indian CGM + EHR, ethics approval

2 **Federated learning**
Train across hospitals; data stays on site

3 **ABDM / FHIR**
Consent manager + FHIR R4 (export ready)

4 **Silent trial → CDSCO**
Prospective validation, then SaMD pathway

Thank you

Code: https://github.com/KoderOP/DigitalTwin · MIT License

Demo: make demo → http://127.0.0.1:8000 (doctor) and /patient/ (patient app)

[FILL IN: team name · members · college / incubator · contact]

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